

Adipic Acid Dihydrazide (ADH) – A Unique Crosslinking Agent and Curative

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Executive Overview

Adipic acid dihydrazide (ADH, CAS No. 1071-93-8) is a symmetrical, difunctional organic compound featuring two terminal hydrazide functional groups (-C(=O)-NH-NH₂) separated by a four-carbon aliphatic adipoyl spacer backbone (-CH₂-CH₂-CH₂-CH₂-). Owing to the nucleophilic reactivity of its primary amine-like hydrazide moieties, ADH functions as an exceptional ambient-temperature crosslinking curative, chain extender, and selective carbonyl scavenger. It demonstrates unique selectivity toward aldehydes and ketones, undergoing rapid condensation to establish stable covalent hydrazone linkages without requiring high activation energy.

Chemical Specifications and Physical Properties

Parameter	Specification / Typical Value
Chemical Name	Adipic Acid Dihydrazide / Hexanedioic acid, dihydrazide
CAS Registry Number	1071-93-8
Empirical Formula	C ₆ H ₁₄ N ₄ O ₂ (Molecular Weight: 174.20 g/mol)
Physical Appearance	White crystalline powder or granules
Active Hydrazide Purity	≥ 99.0% (by titration / HPLC)
Melting Point Range	178 – 182 °C
Water Solubility	Freely soluble in water (~100 mg/mL at 25 °C); insoluble in non-polar organics
Terminal Functionality	Difunctional dihydrazide (-CONHNH ₂ × 2)

Reactivity Mechanism & Selective Scavenging of Formaldehyde

In chemical sensing and formaldehyde scavenging applications, ADH operates via spontaneous nucleophilic addition-elimination with formaldehyde (HCHO). The primary nucleophilic nitrogen atom of the hydrazide attacks the carbonyl carbon of formaldehyde to produce an intermediate carbinolamine, followed by instantaneous dehydration to yield a robust hydrazone bond (-C=N-NH-). When immobilized within a hydrophilic polymer matrix (such as poly(vinyl alcohol) / PVA crosslinked with citric acid), the pendant hydrazide branches of ADH remain freely accessible for binding trace formaldehyde molecules in aqueous or vapor food matrices, facilitating selective analyte capture without cross-reactivity toward competing volatile organic species.

Industrial & Sensor System Applications

1. Formaldehyde Scavenging & Selective Sensing: Incorporating ADH into functionalized electrospun nanofibers creates a high surface-area chemical recognition layer for rapid formaldehyde capture.

2. **Waterborne Coatings & PUD Curatives:** ADH is widely utilized in Diacetone Acrylamide (DAAM) / ADH two-component systems for room-temperature curing of automotive and architectural coatings.
3. **Epoxy Latent Hardener:** Provides high latency and long pot-life at ambient conditions while activating rapidly upon heating above 120 °C.